Section 1.4.2 (pp. 22 – 23)

1. The following lists the functions from highest to lowest order. Those of the same order are shown on one line.

$n!$
2^n, 2^{n-1}
$(3/2)^n$
$n^3 + \lg n$, n^3, $n - n^2 + 5n^3$
n^2
$n \lg n$
n
$\sqrt{n}$
$(\lg n)^2$
$\lg n$
$\lg \lg n$
6

2.
a. $g = O(f)$
b. $f = O(g)$
c. $f = O(g)$
d. $f = O(g)$
e. $f = O(g)$
f. $g = O(f)$
g. $g = O(f)$
h. $f = O(g)$

7)

a. $\sum_{i=1}^{N}(3i+7) = 3*\frac{N*(N+1)}{2}+7N$

$= \frac{3N^2+17N}{2}$

b. $\sum_{i=1}^{N}\left(i^2-2i\right) = \frac{N*(N+1)*(2N+1)}{6} - 2\frac{N*(N+1)}{2}$

$= \frac{2N^3-3N^2-5N}{6}$

c. $\sum_{i=7}^{N} i = \sum_{i=1}^{N} i - \sum_{i=1}^{6} i$

$= \frac{N*(N+1)}{2} - 21$

d. $\sum_{i=5}^{N}\left(2i^2+1\right) = 2\sum_{i=1}^{N} i^2 + \sum_{i=1}^{N} 1 - 2\sum_{i=1}^{4} i^2 - \sum_{i=1}^{4} 1$

$= 2\frac{N*(N+1)*(2N+1)}{6} + N + - 30 - 4$

$= \frac{2N^3+3N^2+4N}{3} - 34$

e. $\sum_{i=1}^{N} 6^i$ $\sum_{i=0}^{N} 6^i - 6^0$

$= \frac{6^{N+1}-1}{5} - 1$

f. $\sum_{i=7}^{N} 4^i = \sum_{i=0}^{N} 4^i - \sum_{i=0}^{6} 4^i$

$= \frac{4^{N+1}-1}{3} - 5461$

$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$